

## Subject Description Form

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| <b>Subject Code</b>                            | EEE512                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Subject Title</b>                           | Quantum Optics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Credit Value</b>                            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Level</b>                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Pre-requisite / Co-requisite/ Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Objectives</b>                              | This subject provides a comprehensive exploration of the principles, theories, and applications of quantum optics. Students will delve into the interactions between light and matter in the quantum regime, covering topics ranging from the basics of quantum mechanics to advanced concepts in quantum field theory and practical applications in quantum technologies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Intended Learning Outcomes</b>              | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>Demonstrate a thorough understanding of quantum optics principles and concepts.</li> <li>Analyze and solve problems related to quantum states of light and matter.</li> <li>Apply quantum mechanics principles to describe atom-field interactions.</li> <li>Design and analyze experiments involving nonclassical light and quantum systems.</li> <li>Explain applications of quantum optics in quantum technologies.</li> <li>Critically evaluate and discuss current research in the field of quantum optics.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Subject Synopsis/ Indicative Syllabus</b>   | <ol style="list-style-type: none"> <li><b>1. Introduction to Quantum Optics</b> <ol style="list-style-type: none"> <li>1-1 Historical overview and significance of quantum optics</li> <li>1-2 Wave-particle duality and quantization of electromagnetic fields</li> <li>1-3 Quantum states of light: photon number states, coherent states</li> </ol> </li> <li><b>2. Quantum Mechanics Review for Quantum Optics</b> <ol style="list-style-type: none"> <li>2-1 Brief review of quantum mechanics principles</li> <li>2-2 Operators, observables, and wave functions in quantum optics context</li> <li>2-3 Commutation relations and uncertainty principle</li> </ol> </li> <li><b>3. Quantum Harmonic Oscillators</b> <ol style="list-style-type: none"> <li>3-1 Quantum harmonic oscillator model</li> <li>3-2 Vibrational modes of atoms and molecules</li> <li>3-3 Energy quantization and ladder operators</li> </ol> </li> <li><b>4. Quantization of Electromagnetic Fields</b> <ol style="list-style-type: none"> <li>4-1 Quantization of the electromagnetic field and photon creation/annihilation operators</li> <li>4-2 Photon statistics and photon number distribution</li> </ol> </li> </ol> |

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|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|---|---|---|---|---|
|                                                                 | <div><div><div>5. Coherent and Squeezed States</div><div>5-1 Properties and generation of coherent states</div><div>5-2 Squeezing of quantum states: photon and quadrature squeezing</div><div>5-3 Applications of squeezed states in quantum optics</div></div><div><div>6. Atom-Field Interactions</div><div>6-1 Interaction of quantum systems with quantized electromagnetic fields</div><div>6-2 Rabi oscillations and Jaynes-Cummings model</div><div>6-3 Spontaneous emission and vacuum Rabi splitting</div></div><div><div>7. Nonclassical Light</div><div>8-1 Nonclassical properties of light: antibunching, sub-Poissonian statistics</div><div>8-2 Hong-Ou-Mandel interference and photon indistinguishability</div><div>8-3 Quantum interference phenomena</div></div><div><div>8. Quantum Optics Applications</div><div>10-1 Quantum cryptography and quantum key distribution</div><div>10-2 Quantum-enhanced sensing and imaging</div><div>10-3 Quantum communication protocols with photons</div></div><div><div>9. Experimental Techniques in Quantum Optics</div><div>11-1 Techniques for generating and manipulating quantum states of light</div><div>11-2 Photon detection and measurement methods</div><div>11-3 Trapped ions and cavity QED experiments</div></div><div><div>10. Future Trends and Research in Quantum Optics</div><div>12-1 Emerging topics in quantum optics research</div><div>12-2 Quantum technologies and their impact on various fields</div><div>12-3 Critical analysis of recent research papers</div></div></div> |             |                                                   |   |   |   |   |   |
| Teaching/Learning Methodology                                   | <div><div>Lecture: The fundamentals of quantum optics will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance.</div><div>Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                                   |   |   |   |   |   |
| Assessment Methods in Alignment with Intended Learning Outcomes | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | % weighting | Intended subject learning outcomes to be assessed |   |   |   |   |   |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             | a                                                 | b | c | d | e | f |
|                                                                 | 1. Continuous Assessments (Assignments and tests)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 40%         | √                                                 | √ | √ | √ | √ | √ |
|                                                                 | 2. Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 60%         | √                                                 | √ | √ | √ | √ | √ |
|                                                                 | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 100 %       |                                                   |   |   |   |   |   |

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|                                      | <p><b>Continuous assessments</b> consist of assignments and mid-term test. These will assess the students' understanding of basic concepts and principles in quantum communication technologies.</p> <p><b>The examination</b> will be conducted to make a comprehensive assessment of students' intended learning outcomes as stated above.</p>                                                                                                                                                                                                                |          |
| <b>Student Study Effort Expected</b> | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|                                      | ▪ Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 33 Hrs.  |
|                                      | ▪ Tutorial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6 Hrs    |
|                                      | Other student study efforts:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
|                                      | ▪ Reading and self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 81 Hrs.  |
|                                      | The total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 120 Hrs. |
| <b>Reading List and References</b>   | <p><b>Text Book</b></p> <ul style="list-style-type: none"> <li>Agarwal, Girish S. Quantum optics. Cambridge University Press, 2012.</li> </ul> <p><b>Reference Book</b></p> <ul style="list-style-type: none"> <li>Lambropoulos, Peter. Fundamentals of quantum optics and quantum information. Springer, 2007.</li> <li>Perina, Jan, Zdeněk Hradil, and Branislav Jurco. Quantum optics and fundamentals of physics, Vol. 63. Springer Science &amp; Business Media, 2012.</li> <li>Duarte, Frank J. Quantum optics for engineers. CRC Press, 2013.</li> </ul> |          |